

7 Limitations and Future Work

The proposed SV-Detect has several limitations. First, although it generalizes well across domains, source models, attack families, and machine-editing settings, it still depends on a probe LLM whose representation geometry affects performance. Our backbone ablations in Section C show broadly consistent behavior across LM backbones, though some variation in transfer robustness remains. Second, while the method is much cheaper than perturbation-based detectors, it still requires a full LLM forward pass and is therefore less efficient than small encoder-based supervised baselines such as RoBERTa. Third, our experiments focus on English benchmarks and on a fixed set of contemporary generation and editing scenarios. Although MIRAGE and PADBen cover important forms of machine-assisted writing, including polishing, rewriting, paraphrasing, laundering, and human-edited AI-generated text, they do not fully capture the broader spectrum of mixed human–AI authorship. In practice, many documents may contain only localized AI-written passages, repeated human edits over AI drafts, or iterative collaboration between a human writer and an LLM. SV-Detect currently produces a document-level score and does not explicitly attribute which spans are human-written, AI-generated, or jointly edited. Finer-grained detection and attribution of mixed authorship therefore remain important directions for future work. Finally, SV-Detect remains a supervised detector. Although it generalizes better than standard text classifiers in our experiments, it still relies on labeled human/machine examples and may require periodic recalibration as generators, prompting practices, editing tools, and human writing styles evolve. We do not expect recalibration frequency to be fixed in advance: it should depend on the deployment domain and be guided by held-out validation data from recent human and AI-assisted writing. Since the probe LLM remains frozen, such adaptation does not require fine-tuning the backbone; it only requires recomputing the layer-wise directions and retraining the lightweight classifier on updated examples.

These limitations point to several directions for future work. One natural extension is to study SV-Detect in multilingual and cross-lingual settings, where both the stylistic signal and the hidden-state geometry may differ substantially. It would also be valuable to extend SV-Detect from document-

level classification to span-level or passage-level attribution for mixed human–AI writing. Another important direction is to develop practical recalibration protocols: for example, periodically refreshing the human/AI calibration set, monitoring validation performance on recent writing samples, and updating the directions when performance under distribution shift begins to degrade. It would also be valuable to explore whether steering vector representations can be combined with lighter-weight encoders to improve efficiency without sacrificing robustness. More broadly, our results suggest that representation space detection is a promising interface between robustness, interpretability, and model internals, and future work could study whether similar ideas apply to other modalities or to broader forms of synthetic content detection. Future work could also compare LogReg-based/PCA/mean-difference steering to other supervised linear projections such as shrinkage LDA or contrastive objectives.

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